

The software for NA62 is split into three main C++ packages: NA62MC, NA62Reconstruction and NA62Analysis. NA62MC is a GEANT4 based Monte Carlo simulation package that simulates the NA62 experiment including the detector geometry and the full simulation of kaon decays, used as a comparison to the data collected by the experiment. NA62Reconstruction is a package that is used to reconstruct either the Monte Carlo data or collected data from the experiment for all detectors. NA62Analysis is used by the physicists to analyse the data produced from the reconstruction to produce all the physics studies that are conducted at NA62.

The NA62 software was designed for use only with the NA62 experiment resulting in minimal configurability and flexibility, with the only option for users to change the software being to enable/disable sub-detectors. With the experiment foreseen to run up to CERN's Long shutdown 3 (LS3), an upgraded software package was needed to allow for the development of new detectors for the future kaon experiments. This software had to be flexible and configurable to allow for the simultaneous use of the software for NA62 data taking and physics analysis, as well as the development of new experiments.

0.1 FlexibleMC

For the implementation of a flexible framework all detector classes were refactored to inherit from a common class which contained the basic functions and variables for all detectors allowing for all classes to be parsed using the same machinery. The input to the simulation was then changed to allow for the configuration of all the detectors to be read in from a configuration file which contained any variables required for individual detectors and the X,Y,Z and rotation of the detectors. An

example entry in the configuration file is shown in Figure 1 where a name was given for the station "STRAW0", along with chamber ID, chmaber type which defines the size of the straws and the chamber geometry which defines if the beam hole is symetric or not. The variables for each subdetector are defined in Table

Name	Arguments	X	Y	Z	θ_x	θ_y	θ_z
Spectrometer("STRAW0",	0, 0, 0)	0	0	183705.9	0	0	0

Figure 1: Example configuration for the STRAW0

Detector Name	Variables
STRAW	Chamber ID: 0,1,2,3 Chamber Type: 0(10mm), 1(5mm) Chamber Mode: 0(default), 1(symmetric)
Spectrometer Magnet	Magnetic field intensity
Vessel	Length InnerRadius OuterRadius Type("BlueTube", "BlueTubeNoRails", "GreyTube")

Table 1

When built inside the simulation, each detector was now contained within its own separate volume called the "Responsibility Region". This was a simple task for most detectors apart from the STRAWs and LAV which were made up of multiple stations that were not continuous, resulting in multiple responsibility regions for each station.

After the simulation was upgraded all the geometries of each detector had to be

validated that they were still correct with respect to the previous version. This was due to the complex nature of refactoring the detectors and that a number volumes had been changed. The validation was completed using the GEANT4 , which is a virtual particle present in the simulation package that will travel in a straight line and interact with all volumes present in the simulation. It was then possible to produce a large square beam of these particles and save every time the particle passed through a boundary between two volumes. Using this it was possible to check the placement and geometry of each detector to mm precision to ensure that the simulation was still accurate. An example of the geantino hits on a single view of STRAW chamber 1 is shown in Figure 2 where it is possible to see the features of the view. For the full validation a larger sample of Geantinos is used to get the required precision, however when there was a discrepancy they were sent in a much smaller area.

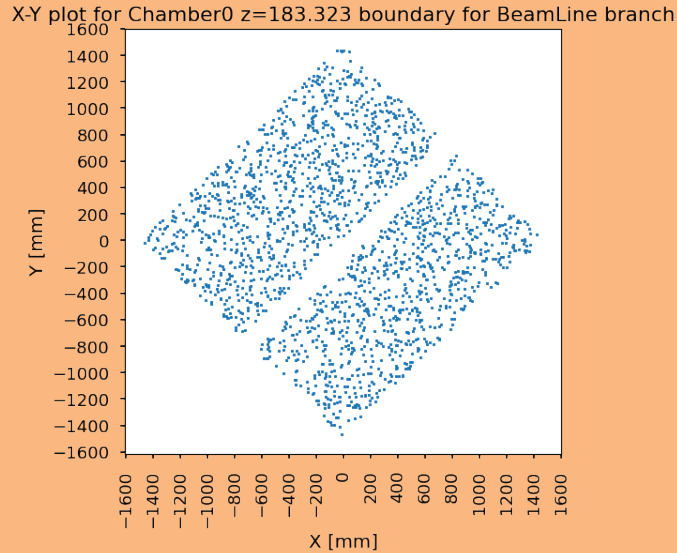
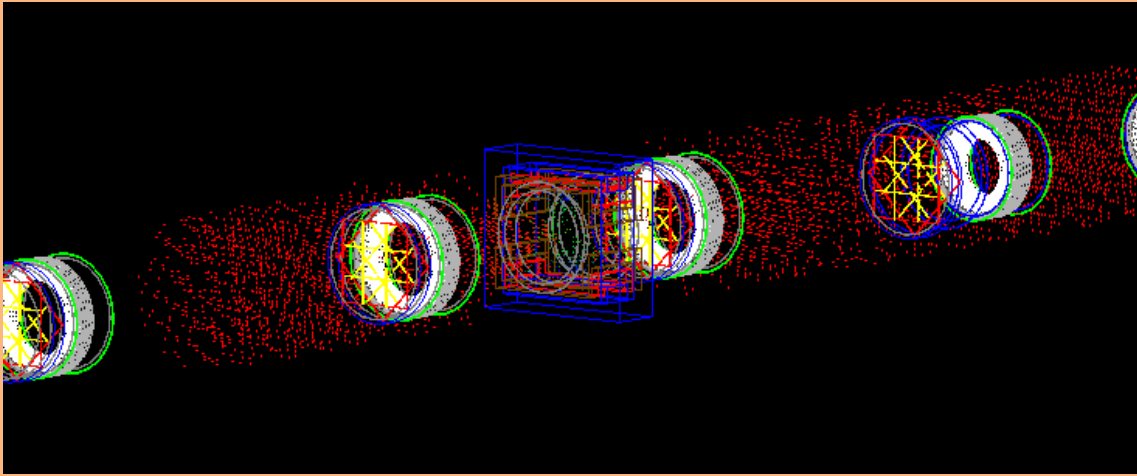


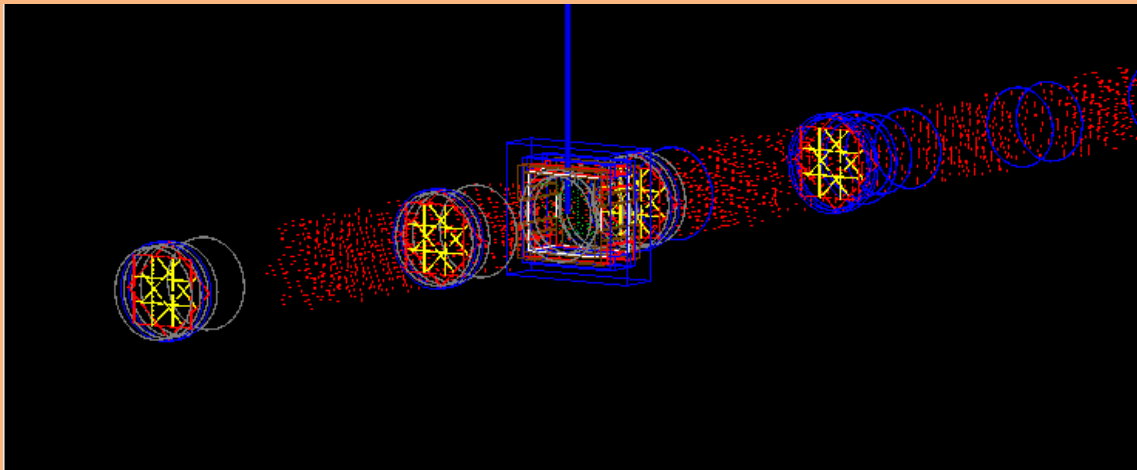
Figure 2: Geantino hits on a single view in straw chamber 1

To reduce the chance of overlaps and to clean up the simulation, a campaign to reduce the size of the unused space in all sub-detectors. The outer volume (Respon-

sibility Region) for all detectors were reduced to just slightly larger than the physical detector which is possible to see in figure 3, as the magnetic fields are calculated in the all volumes not the world volume. In 3(a) the field was calculated outside of the actual physical sub-detector due to the larger outer volume, whereas in 3(b) the volume has been reduced to a few mm's bigger than the actual physical detector and as a result the field is calculated in the correct region.



(a) Previous NA62MC Magnetic field calculations



(b) Upgraded NA62MC Magnetic field calculations

Figure 3: Visualisation from NA62MC of the magnetic field in the spectrometer region

The magnetic field in the simulation is based of real-world measurements of the magnetic field at NA62 using hall probes [1] with information saved in data files which is taken in and used to calulcate the magnetic field as various points in the simulation. In Na62MC the magnetic field is split into three distinct regions; Blutube region which is the magnetic field within the decay region which is used for small corrects, fringe field which are the edges of the MNP33 magnetic field and the MNP33 field which is the main magnetic field in the simulation. In the previous version of the simulation the position of the magnetic fields were hard coded, however in the upgraded version it was possible to move the MNP33 magnet to any position and the magnetic field would be calculated from that point. An example of this flexibility is shown in Figure 4 where the MNP33 magnet was moved in front of the straw chamber 1 and the magnetic field was recalculated from that point within the constructed volumes.

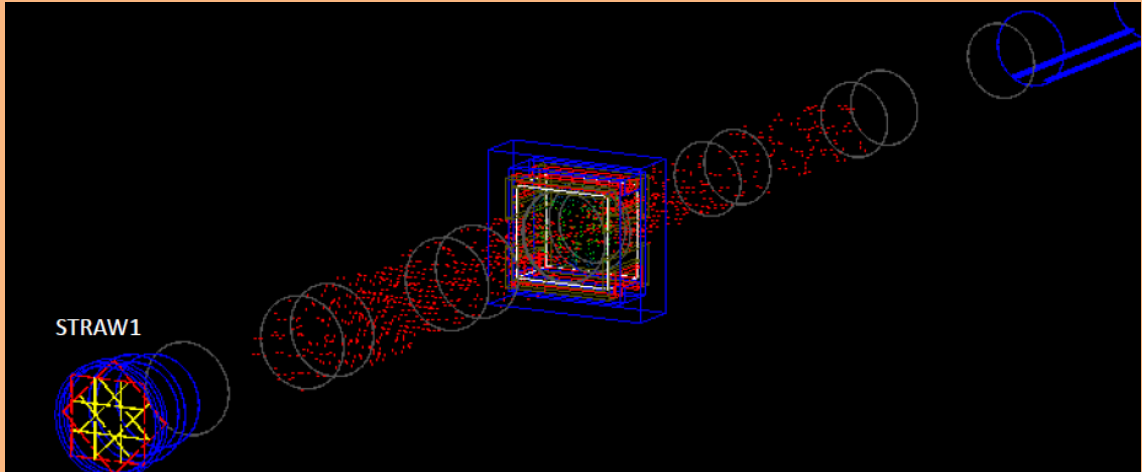


Figure 4: Visualisation from NA62MC with the MNP33 magnet moved in front of the straw chamber 1

Similar to the geometry the modification of the magnetic fields required validation, which was done using similar machinery but with a charged Geantino, similar

to a normal Geantino but interact with magnetic fields.

0.2 SlimMC

Bibliography

- [1] J. R. Fry, G. Ruggiero, and F. Bergsma, “Precision magnetic field mapping for CERN experiment NA62,” *Journal of Physics G: Nuclear and Particle Physics*, vol. 43, no. 12, p. 125004, nov 2016. [Online]. Available: <https://doi.org/10.1088/0954-3899/43/12/125004>